MACEO D. RICHARDS
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EDUCATION -

Kempner Institute at Harvard University:
Post-Baccalaureate Scholar | GPA: 4.00 / 4.00 **July 2024 – June 2026**
Coursework: Systems Neuroscience

Worcester Polytechnic Institute (WPI):
Master of Science, Neuroscience | GPA: 3.87 / 4.00 **May 2024**
Bachelor of Science, Data Science | GPA: 3.75 / 4.00 **May 2024**
Graduated with High Distinction

Coursework: (*graduate level)
Computational Neuroscience*, Math Tools for Neuroscience Data Analysis*, Bioinformatics*, Experimental Design in Social Neuroscience*, Data Analytics and Statistical Learning*, Computational Data Intelligence, Modeling and Data Analysis, Machine Learning, Artificial Intelligence*, Numerical Methods for Linear and Nonlinear Systems

SKILLS -

Technical: Python (MNE, PyTorch, Pyro, TensorFlow, Scikit-Learn, Scipy), MATLAB, R, Linux, Bash, Slurm
Applications: Neural Encoding / Decoding, Information Theory, Bayesian Inference, Reinforcement Learning
Language: Spanish – Proficient (7+ years of foreign language classes)

RESEARCH -

Dendritic Modeling in Artificial Neural Networks – Kempner Institute (Sabatini Lab) **Aug 2024 – Present**
- Designed a dendritic ANN with shunting inhibition to explore tenets of biological computation with deep learning
- Information analyses reveal how locally projecting interneurons modulate excitation-inhibition coordination
- Demonstrated that shunting inhibition enables neurons to stabilize responses across input noise conditions
- Ablation analyses suggest input noise and shunting inhibition govern synapse functional organization in dendrites
- *Dendritic Signal Integration Enables Robust Representation Learning Across Regimes of Input Statistics* **(In Prep)**

Local Learning Rules for Artificial Neural Networks – Kempner Institute (Sabatini Lab) **Oct 2024 – Present**
- Developed biologically plausible learning rules to replace backpropagation for training deep neural networks
- Implemented rule using noise-correlation with neural manifolds to approximate Jacobians from outputs to layers
- Leveraged BrainScore benchmarks to compare ANN layer activity to primate data (V1, V4, IT, behavioral)
- Evaluated benchmarks across learning rules and ANN layers to quantify biological realism of learning dynamics

M.S. Thesis Research, Direct Discriminative Decoder (DDD) – WPI (Yousefi Lab) **Jan 2023 – May 2024**
- Studied spatial coding in the hippocampus with a novel decoder designed for large-scale high-dimensional data
- Integrated traditional time-series models with deep learning in a novel MCMC filter algorithm for neural decoding
- Decoded latent rat trajectory from hippocampal recordings to identify neural representations of spatial location
- Defended thesis research and outlined future methods for detecting sharp wave-ripples and replay events

Facilitation-Depression Recurrent Neural Network – WPI (Yousefi Lab) **Jan 2024 – Apr 2024**
- Emulated presynaptic short-term facilitation and depression dynamics of spiking neurons within a novel RNN
- Proposed the network as a computational paradigm for studying synaptic heterogeneity in biological neurons
- Interpreted resonance frequencies as indicators of synaptic tendency to promote facilitation or depression
- Reviewed how biologically grounded models can inform theory of neural information processing in written report

Internship – MIT Lincoln Laboratory (Biological & Chemical Technologies Group) **May 2022 – Oct 2022**
- Investigated pre-symptomatic COVID exposure detection by training DNN models on biometrics from wearables
- Improved early pathogen detection with transformers and researched interpretability in the time-series domain
- Implemented a multivariate adaptation of LIMESegment to identify biometric indicators of COVID exposure
- Publicly presented strengths and limitations of multivariate LIMESegment with proposed future directions

Interactive Qualifying Project, Ubiquitous EEG/fNIRS Headset – WPI (Yousefi Lab) **Oct 2021 – Dec 2021**

- Designed first iteration of a multimodal EEG–fNIRS recording device for ubiquitous clinical applications
- Addressed the need in clinical studies to simultaneously record multiple modalities in assessing human health
- Conducted sustained attention to response task (SART) for behavioral measures of sustained attention failures
- Correlated EEG features with SART results to identity device-specific neural markers of sustained attention

Summer Project, Introduction to Neural Decoding – WPI (Yousefi Lab) **May 2021 – Aug 2021**

- Created a conceptually accessible demonstration of computational neuroscience for high school students
- Mentored a student in neural decoding to facilitate their learning and receive educational content feedback
- Designed experiments and a decoding pipeline to predict intended screen cursor movement from EEG data
- Delivered presentation to students, the Dean of Arts and Sciences, Neurable CEO and WPI Neuroscience faculty

MANUSCRIPTS -

Richards, M., Safaai, H., Sabatini, B. (In Preparation). *Dendritic Signal Integration Enables Robust Representation Learning Across Regimes of Input Statistics*.

Richards, M., Yousefi, A. (2024). *Dynamical Decoder Model for High-Dimensional Neural Data* [MS Thesis]. WPI.

Richards, M. (2023). *Interpretability of Deep Neural Networks for Pre-Symptomatic Pathogen Exposure Detection*. WPI.

Lam, P.*, **Richards, M.***, Sullivan, J.*, Stevens, S.* (2022). *Toward Wearable Multimodal Neuroimaging*. WPI. *Equal contribution.

POSTERS / ABSTRACTS -

Richards, M., Safaai, H., Sabatini, B. (2026). *Studying the Role of Shunting Inhibition and Dendritic Morphology in Neural Computations* [Submitted Abstract]. Computational and Systems Neuroscience Conference.

Richards, M., Safaai, H., Sabatini, B. (2025). *Dendritic Processing in Artificial Neural Networks based on Shunting Inhibition* [Abstract, Poster]. Frontiers in NeuroAI Symposium, Kempner Institute at Harvard University.

Richards, M., Safaai, H., Sabatini, B. (2025). *Dendritic Processing of Excitation and Shunting Inhibition in Artificial Neural Networks* [Abstract, Poster]. Dendrites: Molecules, Structure and Function Gordon Research Conference.

Krikorian, S., **Richards, M.**, Ziaei, N., Frank, L., Eden, U., Yousefi, A. (2024). *A Novel Dynamic Deep Decoder Model for High-Dimensional Neural Data* [Abstract]. IEEE Engineering in Medicine and Biology Conference.

Richards, M., Ziaei, N., Yousefi, A. (2024). *Decoder Framework for Real-Time Analysis of High-Dimensional Neural Data* [Abstract]. Computational and Systems Neuroscience Conference.

COMMUNITY INVOLVEMENTS -

Kempner Editorial Board (Unpaid), Kempner Institute **Aug 2025 – Present**

- Review content for publication in Kempner news articles and other online materials to ensure scientific accuracy

Co-Admin for Lunch and Learn Event Series (Unpaid), Kempner Institute **Aug 2024 – Present**

- Create an informal setting where participants can explore new topics together and practice public speaking

Peer Well-Being Ambassador (Paid), WPI Center for Well-Being **Aug 2023 – May 2024**

- Raised mental health awareness in the WPI community through peer mentoring and by facilitating events

Graduate Student Peer Advisor (Paid), WPI Heebner Career Development Center **Mar 2023 – May 2024**

- Supported students in professional development through one-on-one peer mentoring and by hosting events

Member (Unpaid), WPI Arts & Sciences Student Advisory Council **Aug 2022 – May 2024**

- Worked with the dean of Arts & Sciences to ideate and facilitate events promoting student research